

LEWWANDI VITHANAGE

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Objective

Seeking a challenging position in Artificial Intelligence to leverage my expertise in machine learning, deep learning, and computer vision. Aiming to contribute to innovative projects at the intersection of cutting-edge AI research and practical problem-solving in fields such as computer vision, natural language processing, and predictive modeling.

Experience

Zindhu Business Solutions (Pvt) Ltd

Oct 2024 – Present

Intern Trainee → Associate Software Engineer

- Architected and implemented a multi-tenant SaaS infrastructure platform on AWS, automating EKS cluster provisioning and deployment using Terraform, Helm, and ECR, reducing infrastructure setup time by 90%.
- Developed a real-time vehicle tracking system for a full-stack e-commerce platform using React.js and RESTful APIs.

Education

General Sir John Kotelawala Defense University

Jan 2022 – Dec 2025

BSc (Hons) in Electronics and Telecommunication Engineering

Ratmalana, Sri Lanka

Sirimavo Bandaranaike Vidyalaya

2012 – 2020

GCE Advanced Level - Physical Science Stream

Colombo 7, Sri Lanka

Projects

Multi-Person Whole Body Human Mesh Reconstruction with Texture Mapping

Dec 2025

- **Tools:** PyTorch, PyTorch3D, SMPL-X, Stable Diffusion(SDXL), Metaverse
- **Motivation:** Generating textured human meshes from a monocular image for immersive VR gaming.
- Developing a pipeline using Global Z-Buffer depth-testing algorithm to eliminate "texture bleeding" in dense scenes.
- Fine-tuned an SDXL inpainting model to synthesise realistic textures for occluded regions using custom masks and implemented morphographical texture padding with backface culling for seamless UV surface rendering.

Sentiment Analysis Web Application

May 2025

- **Tools:** Python, Flask, NLTK, Scikit-learn, TF-IDF, Logistic Regression
- **Motivation:** To build a robust multiclass sentiment classifier enabling data-driven insights from text data.
- Optimized Logistic Regression, Linear SVM and Random Forest using GridSearchCV with 5-fold stratified cross-validation, tuning hyperparameters including L1/L2 regularization
- Deployed Flask REST API with real-time prediction endpoint to deliver sentiment classification on user-generated text

Heart Disease Prediction Web Application

Jan 2025

- **Tools:** Python, Pandas, Flask, JSON Web Tokens
- **Motivation:** To provide early risk assessment support for heart diseases using accessible clinical data.
- Developed a full-stack web application to predict heart disease risk by analyzing 11 clinical parameters.
- Implemented secure authentication using JSON Web Tokens for user login and registration.

Voice Assisted AI Chatbot on Raspberry Pi: Embedded AI System

Dec 2023

- **Tools:** Python, Whisper, Google Gemini API, pytttsx3, Raspberry Pi
- **Motivation:** To explore deploying conversational AI systems on resource-constrained embedded hardware.
- Developed a voice-controlled chatbot integrating Whisper-based speech recognition and Gemini LLM on Raspberry Pi.

Skills

Languages & Tools: Python, JavaScript, C/C++, Bash, Git, Linux, MERN Stack, Flask

AI & Computer Vision: PyTorch, PyTorch3D, TensorFlow, Scikit-learn, OpenCV, SMPL-X

Cloud & DevOps: AWS, Docker, Kubernetes, Helm, Terraform, CI/CD (Github Actions)

Gen AI: Diffusion Models(SDXL), Texture Inpainting

Leadership Experience

Electronic, Robotics & Innovation Club-KDU (*Marketing Director*)

2023 – 2024

Toastmasters Club-KDU (*Education Team Member*)

2022 – 2023

Additional Information

Languages: English, Sinhala (Native)

References

Eng (Mrs.) P. N. Karunanayake

Senior Lecturer (Gr II)

Department of Electrical, Electronic and Telecommunication Engineering

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Pradeep Indrajith

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IFS

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